Trust-Region Newton Methods in a Saddle-Dominated Raman Landscape

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This note asks whether second-order optimization should replace the default first-order Raman-suppression workflow. The answer is useful but mostly negative: trust-region Newton methods are excellent diagnostics for curvature, but they did not become the default optimizer for this Raman objective. The main reason is that the local quadratic model often sees strong negative curvature, so cold-start runs collapse before accepting useful steps. After gauge-safe preconditioning and bounded continuation tests, the dispersion preconditioner gave a small localized benefit, but not a broad result. The best practical lesson is that continuation supplies the path; trust-region Newton can refine and diagnose that path, but it does not replace it.

Claim Status

Status: established diagnostic result; closed as a default replacement
The trust-region implementation is scientifically useful and technically honest after the preconditioner and gauge-projection fixes. The evidence does not support using it as the default Raman optimizer. Preconditioning alone does not rescue bad cold starts, and the best dispersion-preconditioned gains were too localized to justify keeping this as an active Raman-suppression branch.

1 Question and Thesis

The baseline optimizer uses L-BFGS: it follows gradients and builds a cheap low-memory curvature approximation. This note studies a more direct second-order question:

Can Newton-like curvature information give better Raman suppression or more reliable optimization?

The motivation is reasonable. If the objective has a locally quadratic shape, then Newton information should help identify the best step. But this project does not see a clean convex bowl. The resolved Hessian spectra around strong optimized points have both positive and negative eigenvalues, so the landscape is saddle-like. That changes the role of Newton methods. They become good

for diagnosing why the optimizer is stuck, but they are not automatically good at finding a better phase mask from a poor starting point.

The current thesis is:

trust-region Newton is a microscope, not the main engine.

It exposes curvature and failure modes clearly. The engine that still matters most is continuation: move through easier problems into harder ones while carrying a good phase profile forward.

Intuition Check

L-BFGS asks, "which downhill direction should I try next?" Trust-region Newton asks, "if I locally approximate the objective by a quadratic bowl, what step should I trust?" The problem is that the local shape is often not a bowl. It is more like a mountain pass: downhill in some directions and uphill in others. In that setting, a Newton step can be mathematically informative but experimentally unreliable unless the starting point is already on a good path.

2 Common Setup

The controlled field is still a phase-shaped input spectrum,

$$u_\omega(0;\phi) = u_{\omega,0}\exp(i\phi(\omega)).$$

After propagation through the fiber, the physical Raman objective is the Raman-band energy fraction

$$J_{\text{phys}}(\phi) = \frac{\sum_{\omega_k\in\mathcal{B}_{\text{Raman}}} |u_{\omega_k}(L;\phi)|^2\,\Delta\omega}{\sum_k |u_{\omega_k}(L;\phi)|^2\,\Delta\omega}.$$

The propagation model follows the usual generalized nonlinear Schrodinger equation simulation stack for nonlinear fiber optics [1, 2, 3]. The baseline first-order optimizer is L-BFGS [4, 6]. This note keeps the same physics objective but changes the local optimization method.

Intuition Check

The physics problem did not change. We are still trying to find a spectral phase that keeps energy out of the Raman-shifted band. The only new question is whether using Hessian-vector products makes the optimizer smarter.

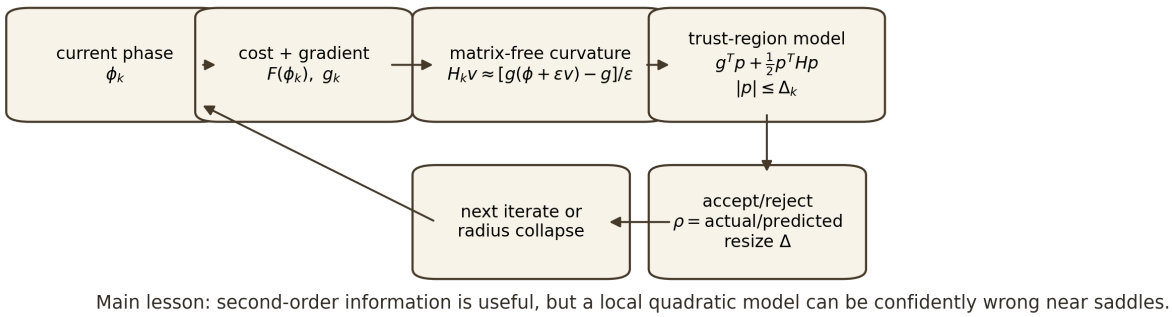

Figure 1: Trust-region workflow. The optimizer builds a local quadratic model from the current cost, gradient, and Hessian-vector products. A trial step is accepted only if the real forward simulation agrees well enough with the quadratic prediction.

3 Trust-Region Math

Let $F(\phi)$ be the scalar objective seen by the optimizer. At iteration k, define

$$g_k = \nabla F(\phi_k), \qquad H_k = \nabla^2 F(\phi_k).$$

A Newton method approximates the objective near ϕ_k by

$$m_k(p) = F(\phi_k) + g_k^T p + \frac{1}{2} p^T H_k p.$$

The trust-region step solves the bounded quadratic problem

$$\min_p \ g_k^T p + \frac{1}{2} p^T H_k p \quad \text{subject to} \quad \|p\|_2 \le \Delta_k,$$

where Δ_k is the radius of the region where we trust the quadratic model.

After proposing p_k, the optimizer compares predicted improvement to real improvement:

$$\rho_k = \frac{F(\phi_k) - F(\phi_k + p_k)}{m_k(0) - m_k(p_k)}.$$

If ρ_k is large enough, the step is accepted. If ρ_k is small or negative, the step is rejected and the trust radius shrinks. The implementation uses the standard trust-region thresholds

$$\rho < 0.25 \Rightarrow \Delta \leftarrow 0.25\|p\|, \qquad \rho > 0.75 \text{ and boundary step} \Rightarrow \Delta \leftarrow \min(2\Delta, \Delta_{\max}).$$

Intuition Check

The trust radius is a safety belt. A Newton model may be accurate very close to the current phase mask but wrong far away. If a trial step improves the real simulation as much as the model predicted, the optimizer trusts a larger neighborhood. If the trial step fails, the optimizer becomes more cautious.

4 Steihaug-CG and Negative Curvature

The full Hessian is too large to build explicitly in the main simulations, so the trust-region subproblem is solved with truncated conjugate gradient, often called Steihaug-CG [5, 6]. This only needs products $H_k v$, not the matrix H_k itself.

The important saddle-specific rule is:

$$d^T H_k d \leq 0 \quad \Rightarrow \quad \text{negative curvature detected.}$$

When this happens, the inner solve stops and moves to the boundary of the trust region in that direction. This is mathematically correct: in a direction with negative curvature, the local quadratic model keeps decreasing as the step grows, so the constrained best step sits on the boundary.

The Hessian-vector product is computed by finite differencing gradients,

$$H(\phi)v \approx \frac{\nabla F(\phi + \epsilon v) - \nabla F(\phi)}{\epsilon},$$

with an adaptive ϵ chosen from the current gradient and vector norms. This is matrix-free but not free: each HVP costs extra gradient work.

Intuition Check

Negative curvature is not a small numerical inconvenience. It means the "bowl" picture is wrong. The surface is bending downward in at least one direction. That is why the cold-start runs can keep proposing boundary steps that look good to the quadratic model but fail in the real simulation.

5 Gauge Projection

Phase-only Raman suppression has directions that do not change the physical Raman energy ratio in the usual way. A constant phase mostly changes global phase, and a linear phase in frequency mostly shifts the pulse in time. These are gauge-like directions. If they are not handled carefully, curvature diagnostics can waste effort on directions that are not meaningful control degrees of freedom.

The optimizer projects gradients, Hessian-vector inputs, Hessian-vector outputs, and preconditioned residuals into the gauge-complement subspace:

$$v_\perp = (I - P_{\mathcal{G}})v, \qquad \mathcal{G} = \text{span}\{1, \omega - \bar{\omega}\}.$$

Here $P_{\mathcal{G}}$ is the least-squares projector onto constant and linear-in-frequency phase directions. A gauge-leak guard checks that accepted steps stay in this projected subspace.

Intuition Check

This is a linear algebra cleanup step. If two phase changes are physically equivalent for the Raman objective, we should not let the optimizer spend its curvature budget distinguishing them. The projection says: remove the boring global-phase/time-shift directions, then optimize the remaining shape.

6 Preconditioning

Preconditioning tries to make the inner CG problem easier by changing how directions are scaled. In the preconditioned solve, the residual r is mapped through an approximate inverse curvature operator,

$$z = M^{-1}r,$$

and the CG search directions are built from z rather than directly from r. If $M \approx H$ and M is stable, the inner solve can require fewer iterations or propose better-scaled steps.

Three preconditioner ideas were tested in bounded cases:

- **none:** identity scaling, the honest baseline.
- **dispersion:** frequency-dependent diagonal scaling $d_i = 1 + \alpha(\omega_i / \max|\omega|)^2$.
- **DCT/reduced:** a low-dimensional cosine-basis Hessian approximation with a stabilizing shift.

The key implementation fix was that preconditioned residuals and directions must be projected back into the gauge-complement space. Pointwise division by a diagonal preconditioner does not commute with the gauge projection, so without re-projection the preconditioner can create illegal gauge leakage.

Intuition Check

Preconditioning is like changing units before doing geometry. If one direction is naturally stiff and another is naturally soft, the optimizer can behave better after rescaling. But rescaling cannot create a good optimization path from nothing. In this project, the starting path mattered more than the preconditioner.

7 Implementation Surface

Component	Role
`core.jl`	Steihaug-CG subproblem solver, trust-radius update, typed exit codes, and subproblem result contract.
`optimize.jl`	Raman trust-region outer loop, matrix-free HVP wrapper, acceptance ratio, radius collapse checks, gauge projection, and standard result object.
`pcg.jl`	Preconditioned CG solver with gauge-safe residual and direction projection.
`preconditioner.jl`	Diagonal power, dispersion, and reduced DCT preconditioner factories.
`telemetry.jl`	Per-iteration records: objective, gradient norm, radius, ρ, accepted flag, HVP counts, and curvature probes.

All listed files are in `scripts/research/trust_region/`.

The result object exposes a `minimizer` field so the normal standard image pipeline can save phase diagnostics and spectral-evolution heat maps.

8 Experimental Strategy

The work was organized into four checks.

1. **Hessian landscape check.** Probe the largest positive and most negative resolved Hessian eigenvalues at optimized reference points. The goal is to test whether the optimizer is near a convex minimum or a saddle-like stationary point.
2. **Cold-start radius sweep.** Start from $\phi = 0$ and vary the initial trust radius Δ_0. If all radii collapse with no accepted steps, then the problem is not simply "bad initial radius."
3. **Gauge-safe preconditioner reruns.** Compare no preconditioner, dispersion scaling, and reduced/DCT scaling after fixing the preconditioner wiring and gauge-projection issue.
4. **Continuation-assisted ladder.** Solve an easier fiber length first, then move the phase mask to harder lengths and compare no preconditioner against dispersion preconditioning.

Intuition Check

This is not one giant benchmark. It is a sequence of sanity checks. First ask what the landscape looks like. Then ask whether trust-region can leave the control point. Then ask whether preconditioning helps. Finally ask whether it helps when continuation has already supplied a reasonable path.

9 Hessian Geometry Result

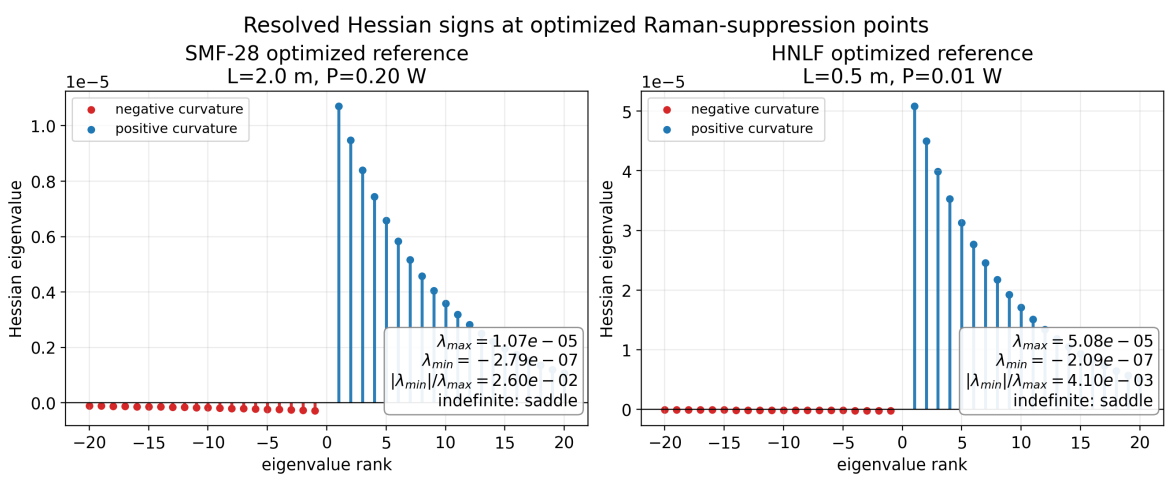

Figure 2: Resolved Hessian signs at two optimized reference points. Both cases show positive and negative curvature after gauge handling, so the local geometry is saddle-like rather than a clean convex minimum.

At the optimized SMF-28 reference point, the largest resolved positive eigenvalue was about 1.07×10^{-5}, while the most negative resolved eigenvalue was about -2.79×10^{-7}. At the optimized HNLF

reference point, the corresponding values were about 5.08×10^{-5} and -2.09×10^{-7}. The negative directions are smaller than the stiff positive directions, but they are not zero.

Intuition Check

The solution is not sitting at the bottom of a smooth bowl. It is more like a high-dimensional pass: some directions curve up, and some directions curve down. That explains why a second-order method can detect a problem even when the first-order gradient is already small.

10 Cold-Start Radius Sweep

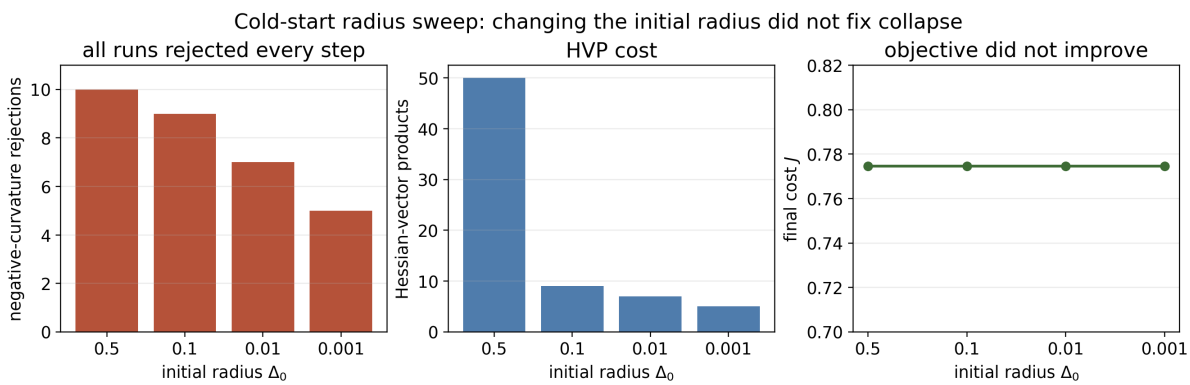

Figure 3: Cold-start trust-region sweep on the canonical SMF-28 case. All initial radii collapsed, no run accepted a step, and the objective stayed at the unoptimized baseline value.

Δ_0	exit	final J	accepted	negative-curvature rejections
0.5	radius collapse	7.746118×10^{-1}	0	10
0.1	radius collapse	7.746118×10^{-1}	0	9
0.01	radius collapse	7.746118×10^{-1}	0	7
0.001	radius collapse	7.746118×10^{-1}	0	5

The largest-radius run also measured strong local indefiniteness during the collapse: $\lambda_{\min} \approx -303.1$ and $\lambda_{\max} \approx 547.4$. The absolute values are on the local objective scale used by that trust-region run, not directly comparable to the reference Hessian plot above. The sign pattern is the important part.

Intuition Check

Changing the initial radius is like asking the optimizer to take a smaller or larger first step. This sweep says the first-step size was not the real issue. The local quadratic model itself was not giving acceptable real improvements from the zero phase mask.

11 Representative Result Pages

Each page below pairs the phase diagnostic with the corresponding spectral evolution heat map. The first page is the no-optimization reference.

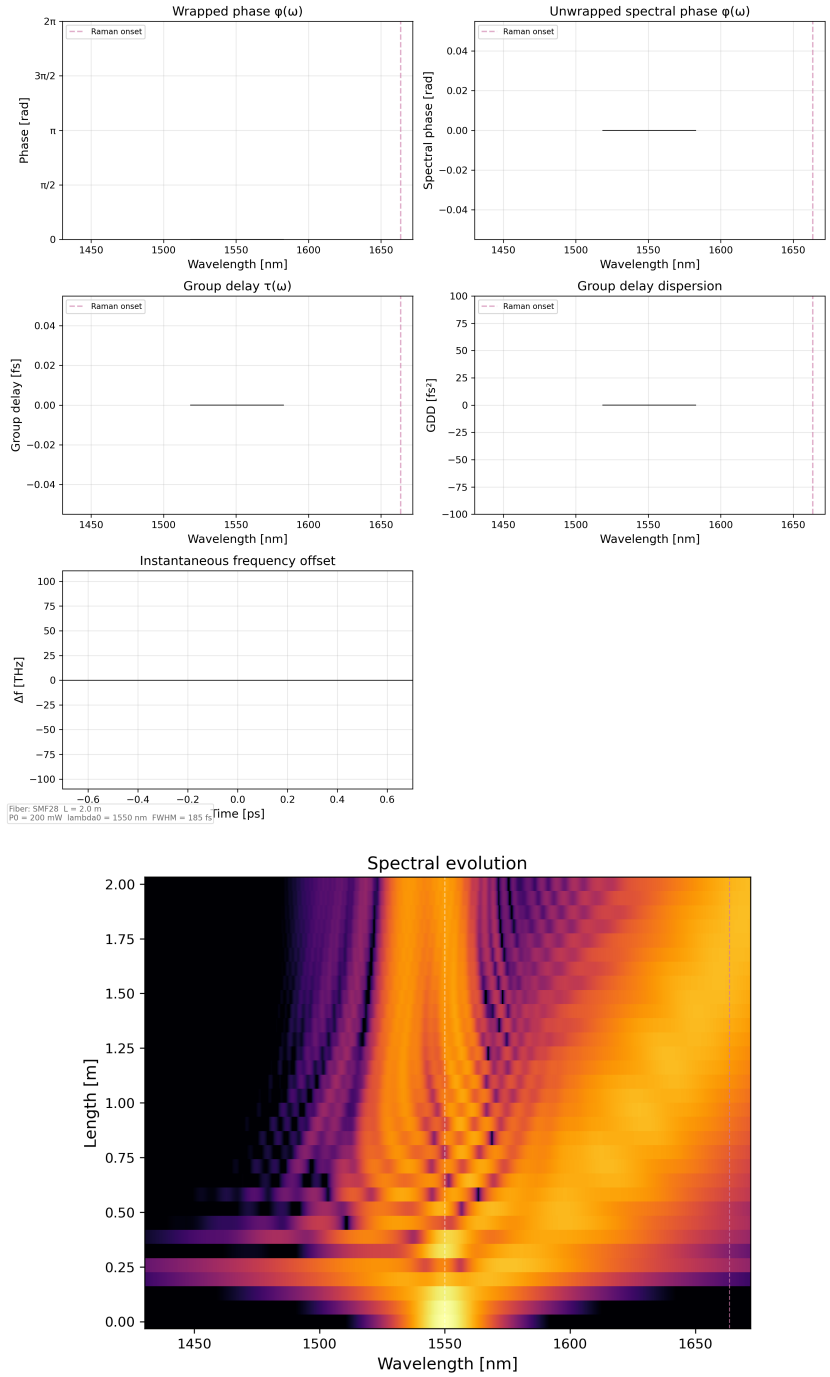

Figure 4: No-optimization control. The phase is flat, and the heat map shows the unshaped spectral evolution. This is the reference behavior that the optimized runs must beat.

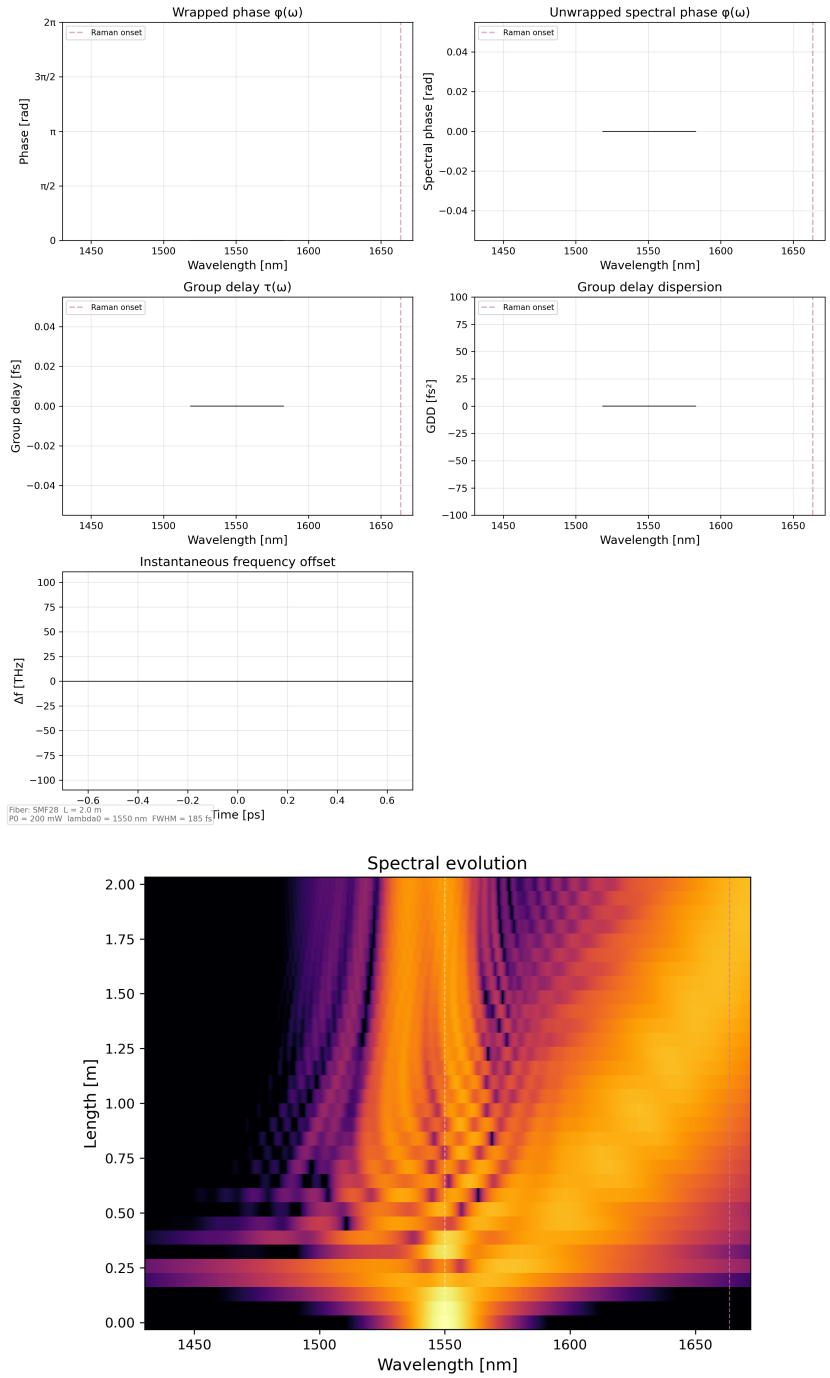

Figure 5: Cold-start trust-region collapse. Since no step was accepted, the phase diagnostic remains essentially the same as the control, which visually matches the numerical result that the objective did not improve.

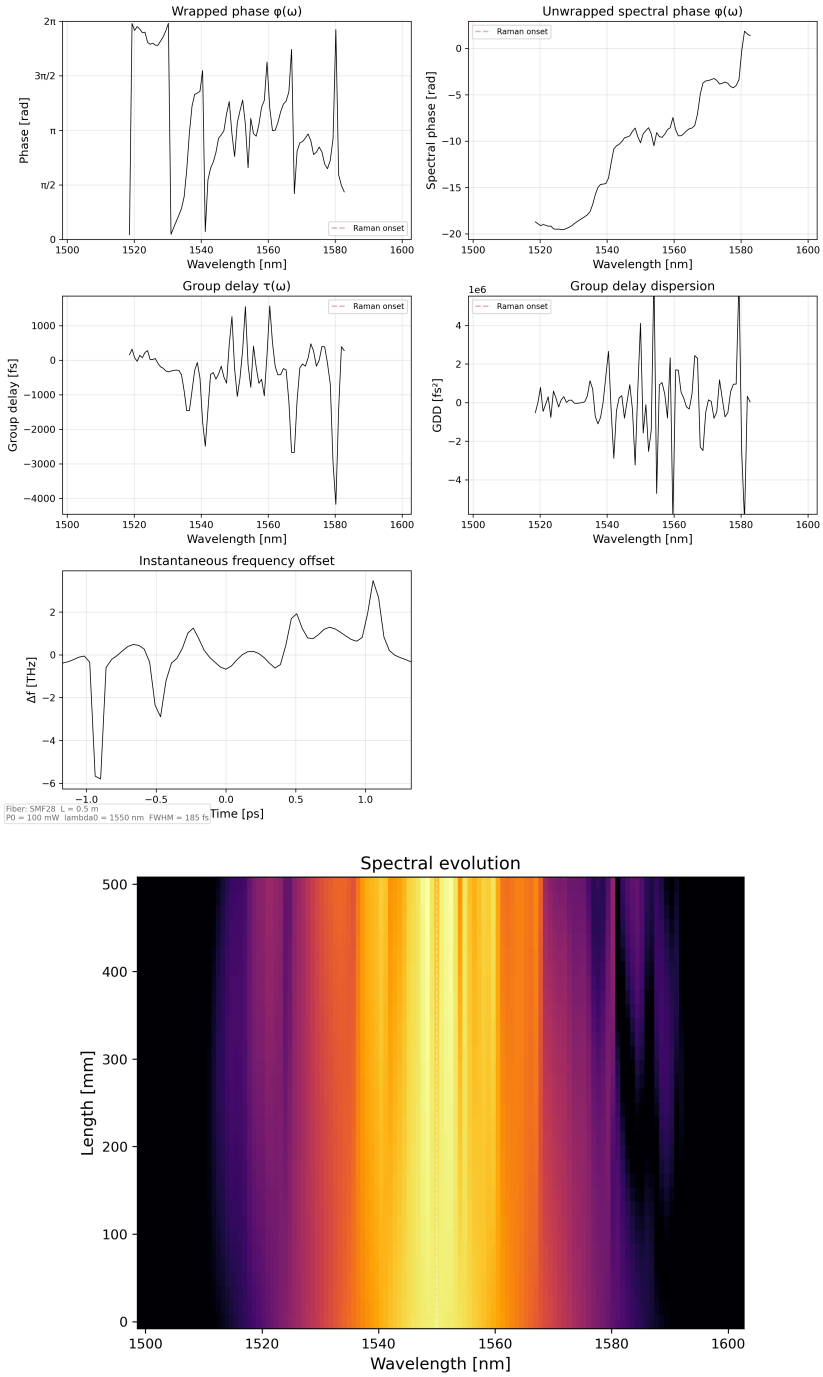

Figure 6: Continuation seed from an easier fiber-length solve. This phase mask is not produced by cold-start trust-region Newton; it is the kind of useful starting point that continuation can supply.

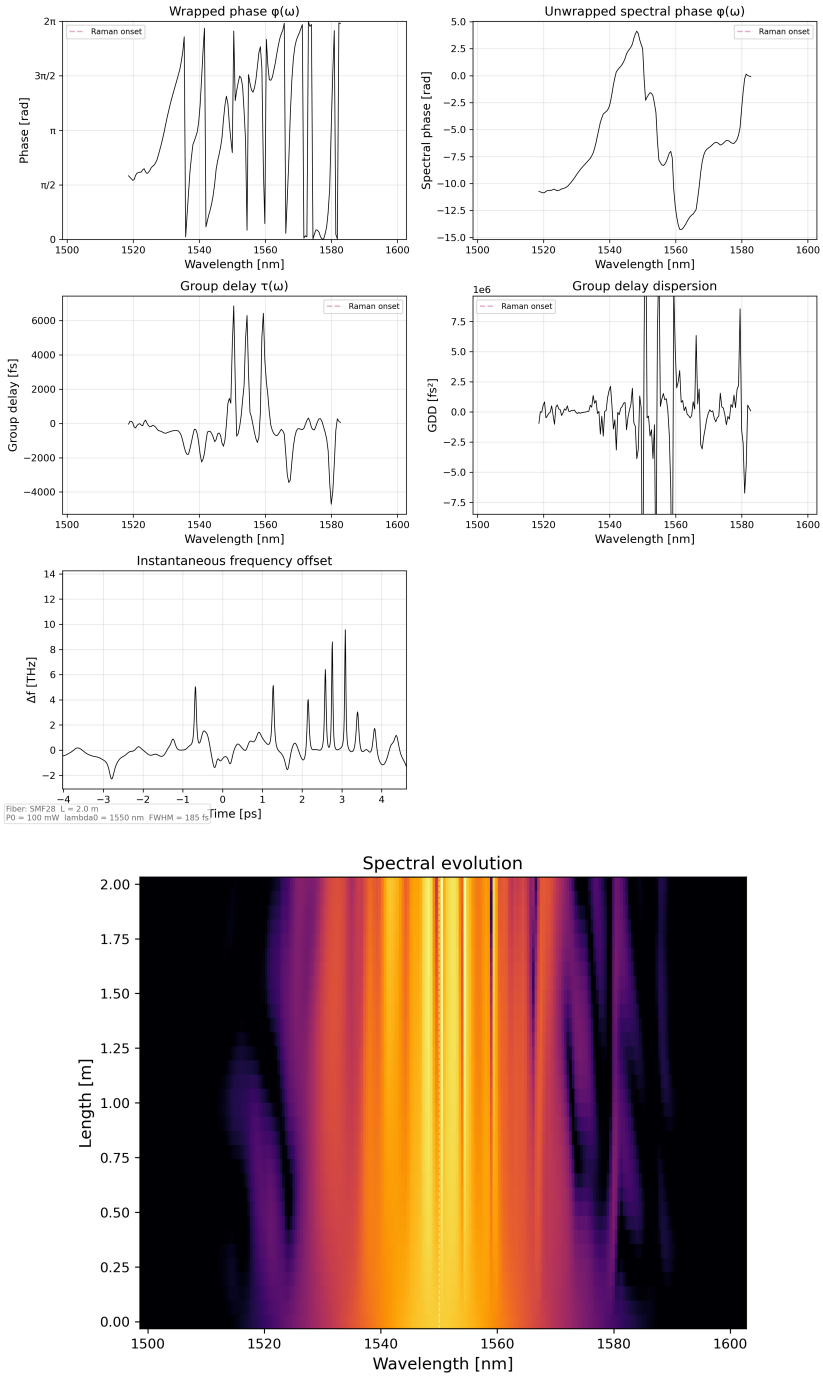

Figure 7: Continuation-assisted trust-region result without preconditioning on the harder ladder rung. This is the honest baseline for the local second-order comparison.

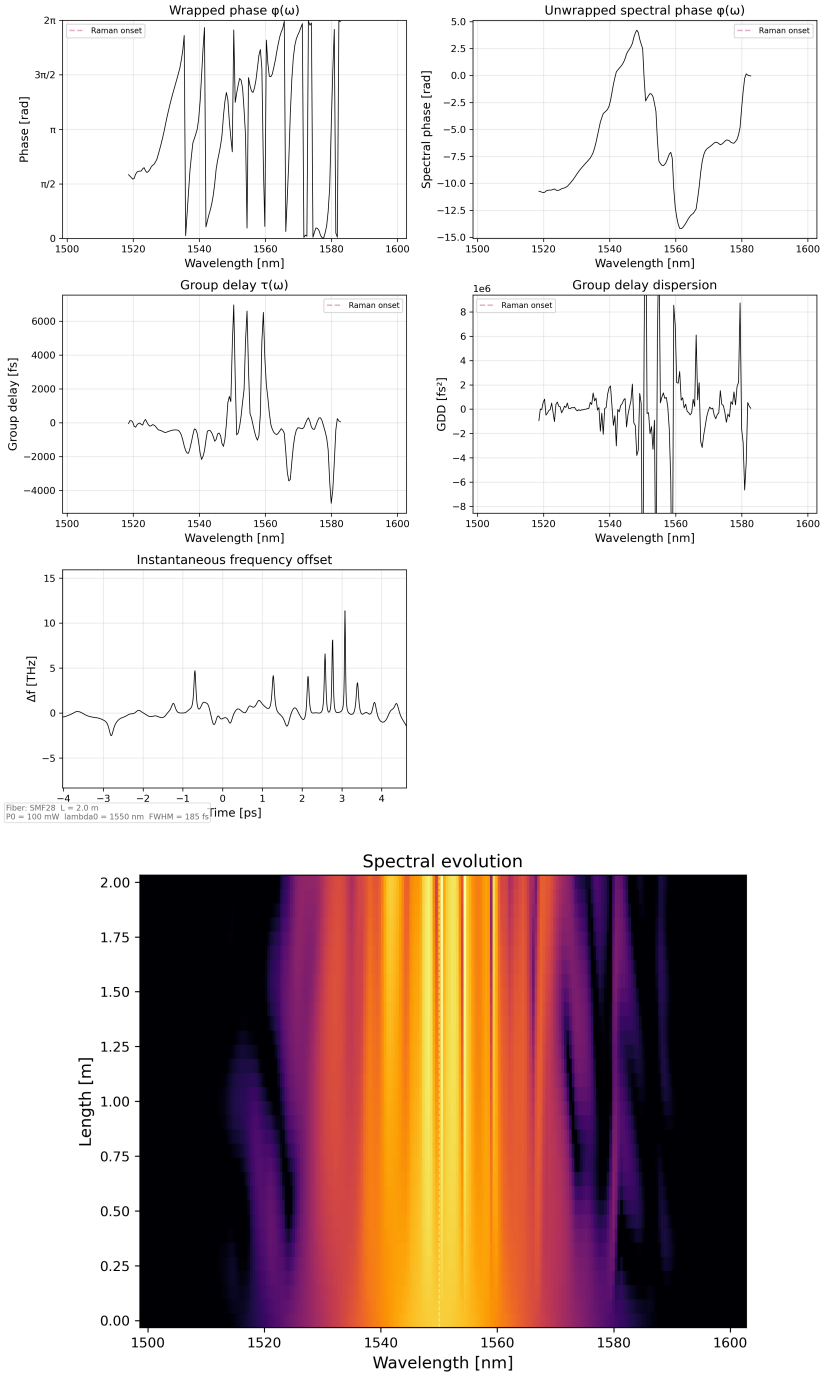

Figure 8: Continuation-assisted trust-region result with dispersion preconditioning on the same harder ladder rung. This was the clearest bounded case where dispersion scaling improved the final objective and accepted-step count.

12 Continuation and Preconditioning Results

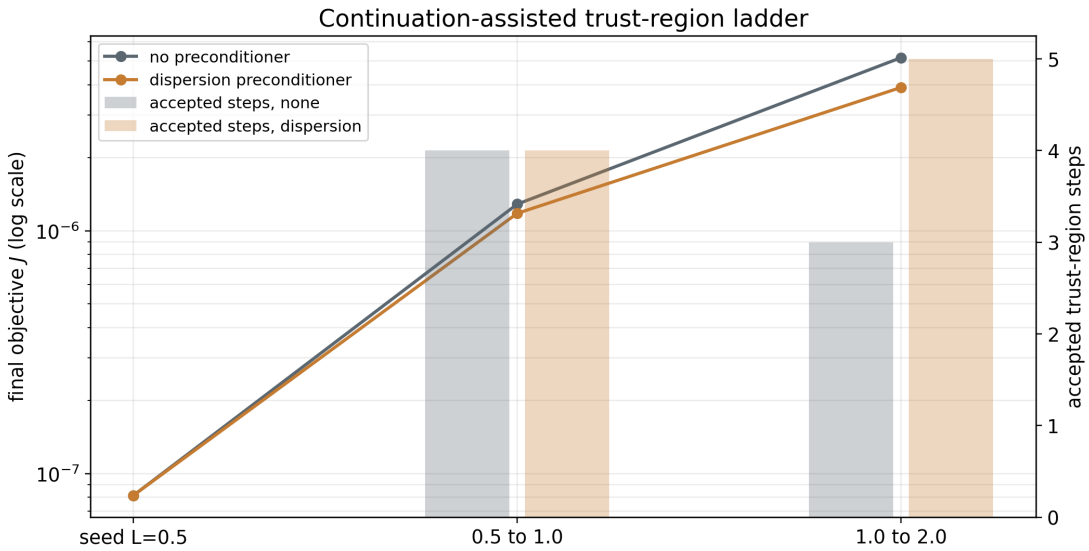

Figure 9: Short continuation ladder at fixed input power. Dispersion preconditioning was modestly better than no preconditioner on the harder rung, but only after continuation supplied a useful starting point.

Step	**method**	**final** J	**accepted**	**exit**
Seed $L = 0.5\,\mathrm{m}$	L-BFGS	8.14×10^{-8}	–	converged
$0.5 \to 1.0\,\mathrm{m}$	no precond.	1.29×10^{-6}	4	1st-order saddle
$0.5 \to 1.0\,\mathrm{m}$	dispersion	1.18×10^{-6}	4	1st-order saddle
$1.0 \to 2.0\,\mathrm{m}$	no precond.	5.17×10^{-6}	3	max iter.
$1.0 \to 2.0\,\mathrm{m}$	dispersion	3.90×10^{-6}	5	max iter.

This is the strongest positive evidence in the trust-region lane. It does not say that dispersion preconditioning solves the Raman problem. It says that, on this continuation-assisted ladder, dispersion preconditioning made the harder local step sequence somewhat better than the identity baseline.

Intuition Check

Continuation got the optimizer into the neighborhood. Dispersion preconditioning helped locally once it was there. That is very different from saying the preconditioner can replace continuation.

13 Power Validation and Closure

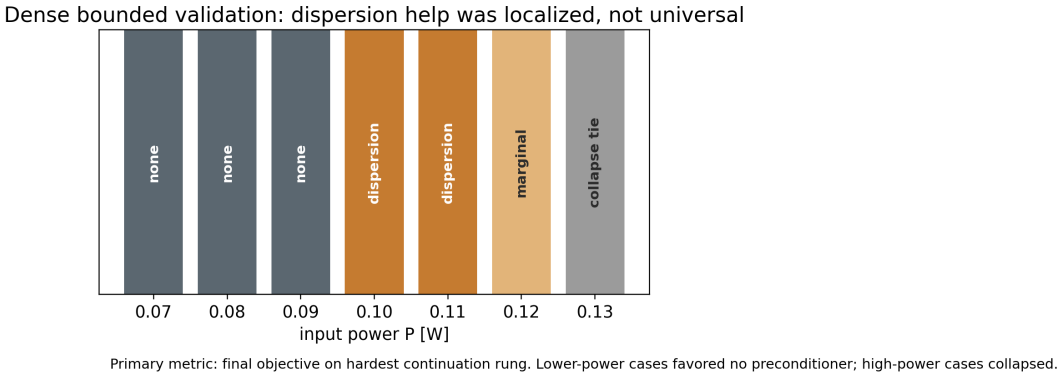

Figure 10: Dense bounded validation across nearby powers. The dispersion preconditioner was not a universal win: the benefit appeared localized near a middle-stress case and disappeared or became weak outside that window.

The follow-up power validation used the hardest continuation rung as the primary metric. Lower-power cases favored no preconditioner. Around $P = 0.10\,\mathrm{W}$, dispersion preconditioning gave the clearest benefit. At higher power, both variants ran into collapse-like behavior, so a tiny dispersion edge there is not a strong scientific result.

That is why the project-level conclusion is closure rather than expansion. The trust-region path taught us a lot about the landscape, but it did not produce a robust replacement for the simpler continuation-plus-L-BFGS strategy.

Intuition Check

A method that wins only in a narrow setting can still be scientifically interesting. But it should not become the default workflow. For a default optimizer, the project needs stable wins across nearby regimes.

14 Presentation Capsule

If this note becomes a second-order-methods slide, the clean message is:

- Trust-region Newton methods use a local quadratic model, but this Raman landscape is strongly saddle-dominated.
- Negative curvature is real evidence, not a plotting artifact: small gradients do not guarantee a simple minimum.
- Cold-start trust-region Newton did not replace L-BFGS or continuation; it mainly explained why naive second-order steps are brittle here.
- The presentation value of this note is diagnostic: it explains failure modes and motivates continuation, preconditioning, and careful trust checks.

15 Lessons and Limitations

Lesson 1: saddle geometry is real. Resolved Hessian spectra show both positive and negative curvature, so a small gradient does not mean a clean minimum.

Lesson 2: cold-start trust-region Newton is brittle here. The radius-sweep result rules out the simplest explanation that the first radius was just chosen badly.

Lesson 3: gauge handling is not optional. Preconditioning can push vectors out of the gauge-complement subspace unless residuals, directions, and final steps are re-projected.

Lesson 4: continuation matters more than preconditioning. Dispersion scaling helped only after a continuation path supplied a useful initial phase mask.

Limitation 1: bounded cases are not final production runs. The preconditioning comparisons were intentionally limited. They are enough to make a research decision, but they are not a comprehensive method paper.

Limitation 2: HVPs are finite-difference objects. The Hessian-vector products depend on gradient accuracy and finite-difference step choice. They are useful, but they are not exact symbolic Hessians.

Limitation 3: objective scale matters. Curvature numbers from different objective conventions should not be mixed as if they live on the same numerical scale. The reliable comparison is the sign pattern and the behavior of the actual trust-region runs.

16 Reproduction Capsule

The trust-region utilities live under `scripts/research/trust_region/`. Substantial reruns should use Julia threading and the project heavy-job workflow when the grid or sweep is large:

```
julia --project=.  -t auto <trust-region-driver>.jl
```

For a complete run, the important outputs are:

- trust report markdown with exit code, final objective, accepted-step counts, and HVP budget;
- telemetry CSV with per-iteration ρ, radius, and step status;
- standard image set for each representative result: phase diagnostic, shaped evolution, and unshaped evolution;
- visual inspection of representative best, typical, failed, and control cases before using the result in a report.

17 Sources and References

References

[1] G. P. Agrawal. *Nonlinear Fiber Optics*, 6th ed. Academic Press, 2019. https://www.sciencedirect.com/book/9780128170427/nonlinear-fiber-optics

[2] J. M. Dudley, G. Genty, and S. Coen. "Supercontinuum generation in photonic crystal fiber." *Reviews of Modern Physics*, 78, 1135–1184, 2006. https://doi.org/10.1103/RevModPhys.78.1135

[3] J. Hult. "A fourth-order Runge-Kutta in the interaction picture method for simulating supercontinuum generation in optical fibers." *Journal of Lightwave Technology*, 25(12), 3770–3775, 2007. https://doi.org/10.1109/JLT.2007.909373

[4] D. C. Liu and J. Nocedal. "On the limited memory BFGS method for large scale optimization." *Mathematical Programming*, 45, 503–528, 1989. https://doi.org/10.1007/BF01589116

[5] T. Steihaug. "The conjugate gradient method and trust regions in large scale optimization." *SIAM Journal on Numerical Analysis*, 20(3), 626–637, 1983. https://doi.org/10.1137/0720042

[6] J. Nocedal and S. J. Wright. *Numerical Optimization*, 2nd ed. Springer, 2006. https://link.springer.com/book/10.1007/978-0-387-40065-5

[7] A. R. Conn, N. I. M. Gould, and P. L. Toint. *Trust Region Methods*. SIAM, 2000. https://doi.org/10.1137/1.9780898719857

[8] J. L. Morales and J. Nocedal. "Automatic preconditioning by limited memory quasi-Newton updating." *SIAM Journal on Optimization*, 10(4), 1079–1096, 2000. https://doi.org/10.1137/S1052623497327854